Discrete Mathematics Exam

2025 - Moed B

Instructions to Candidates:

The exam consists of 10 short questions and 2 open-ended questions. You must answer **all** of them (no choice).

1. Write your solution on the exam sheet.
2. When asked to "explicitly write the set/relation/function," you must list all elements of the set.
3. No reference materials may be used during the exam. A formulas sheet is provided at the end of the document.
4. The time allocated for the exam is three hours.
5. The questions are not arranged in increasing order of difficulty.

Good Luck!

Part A - Short Questions

Write your answers clearly in the marked spaces.

No justification is required unless specifically asked for.

For counting questions, you may write your answer using the standard notation used in the course, except for the 'Σ' symbol and ellipses '...'.

Question 1 (6 points)

Write the numbers of the **true** statements:

1. $\forall x\big(P(x) \wedge Q(x)\big) \equiv \forall x\big(P(x)\big) \wedge \forall x\big(Q(x)\big)$
2. $\forall x\big(P(x) \vee Q(x)\big) \equiv \forall x\big(P(x)\big) \vee \forall x\big(Q(x)\big)$
3. $\exists x\big(P(x) \wedge Q(x)\big) \equiv \exists x\big(P(x)\big) \wedge \exists x\big(Q(x)\big)$
4. $\exists x\big(P(x) \vee Q(x)\big) \equiv \exists x\big(P(x)\big) \vee \exists x\big(Q(x)\big)$

1,4

Question 2 (6 points)

Let $A = \{1,2,3\}$, $B = \{\emptyset, 3,4\}$. Write explicitly, the members of the set

$$\big((A \cup B) \triangle A\big) \times P(A \cap B)$$

$\{(\emptyset, \emptyset), (\emptyset, \{3\}), (4, \emptyset), (4, \{3\})\}$

Question 3 (6 points)

$f: \mathbb{N} \to \mathbb{N}$ *, defined as follows:*

$$f(n) = \begin{cases} n^2 & n \in \mathbb{N}_{even} \\ n^2 + 2 & n \in \mathbb{N}_{odd} \end{cases}$$

Given that $f \circ g = 2n + 1$. *Compute* $g(13)$. *No need to write the explanation of the calculation*

$g(13) = 5$

We can see that if the input of f is odd the output is even and vice versa.

We see that $(f \circ g)(13) = 27$. By definition of composition $f\big(g(13)\big) = 27$.

Denote $g(13) = x$ and get $f(x) = 27$. Since $x^2 + 2 = 27$ we get $x = 5$.

Question 4 (6 points)

Given the digits 1,2,3,…,9. In how many 7-digits numbers, composed from the given numbers, there is no digit that appear exactly twice.

$$9^7 - \binom{9}{1}\binom{7}{2}8^5 + \binom{9}{2}\binom{7}{2}\binom{5}{2}7^3 - \binom{9}{3}\binom{7}{2}\binom{5}{2}\binom{3}{2}6^1$$

We get it by complement of inclusion-exclusion

A_i *– Set of 7-digits number where the digit I appears exactly twice*

Note - Beside S_1, S_2, S_3*,all other* $S_i = 0$

Question 5 (6 points)

Given the digits 1,2,3,4,5. In how many 4-digits numbers, composed from the given numbers, the digit 3 appears more times than the digit 1.

$$\frac{5^4 - \binom{4}{2} - \binom{4}{1}\binom{3}{1}3^2 - 3^4}{2} = 215$$

Using symmetry and complement

We subtract from the total amount, all numbers that 1,3 appears equaly (doesn't appear, appear once, appear twice) and divide it by 2.

Question 6 (6 points)

What is the number of non-negative integer solutions for the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 7$$

such that $x_5 \in \mathbb{N}_{odd}$?

Separate to disjoint cases ($x_5 = 1,3,5,7$)

$$\binom{9}{3} + \binom{7}{3} + \binom{5}{3} + 1 = 130$$

Question 7 (6 points)

Let $A = \{0,1,2,\dots,999\}$ *.Define equivalence realtion* R *on* A:

xRy *iff the set of digits of* x *contained in the set of digits of* y.

e.. $23R302$ *since* $\{2,3\} \subseteq \{0,2,3\}$, *or* $5R555$ *since* $\{5\} \subseteq \{5\}$.

but $\neg(450R454)$ *since* $\{4,5,0\} \nsubseteq \{4,5\}$.

How many minimal elements there are in R? How many maximal elements there are?

$$\binom{10}{1} + \binom{10}{2} + \binom{10}{3}$$

Question 8 (6 points)

Let $n \geq 2$ be an <u>even</u> natural number. Given the expression:

$$\sum_{k=0}^{\frac{n}{2}} \binom{n}{2k}$$

Which of the following is equal to the given expression

Write the number of the expression and give short explanation for your choice (Not necessarily combinatoric explanation).

1. $n!$ **2.** $\binom{2n}{n}$ **3.** 2^{n-1} **4.** $\binom{2n+1}{2}$

3

Sum of all even binomial coefficients, which is half of the sum of all binomial coefficients.

Question 9 (6 points)

Let A be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$

Define equivalence relation on A in the following way:

$$fRg \quad \Leftrightarrow \quad Im(f) = Im(g)$$

Write (without explanation) the equivalence class of $i_{\mathbb{R}}$

All surjective (onto) functions

Explenation:

$[i_{\mathbb{R}}]_R = \{f \in A | Im(f) = Im(i_{\mathbb{R}})\} =$

$= \{f \in A | Im(f) = \mathbb{R}\} =$

$= \{f \in A |\ f\ is\ onto\}$

Question 10 (6 points)

x boys and 7 girls arrived to the PURIM party. Each one of the boys gave a present to at least one of the girls. What is the minimal number x that will assure that at least 2 boys gave present to the exact set of girls? (No need an explanation)

Using PHP:

There are $2^7 - 1 = 127$ *subsets of girls. These are the pigeonholes.*

The boys are the pigeons.

We need at least 128 pigeons, to ensure that there is a pigeonhole with at least 2 pigeons.

Part B - Open Questions

Write your answers in the marked spaces. Full proofs must be provided for your answers.

Be sure to write clearly and legibly! The grade will depend on both the correctness and the clarity of your answer.

Question 11 (20 points)

*Let A be a set, and let R be a **total** order on A.*

Define the relation S on A as:

$$S = \{(x, y) \in A \times A \mid \neg(xRy) \wedge \neg(yRx)\}$$

A. *(10 points) Prove: S contained in the equality relation i_A*

B. *(10 points) prove or disprove: S is reflexive*

א. A. Proof:

Let $(x, y) \in S$ .We prove $x = y$ *and we can conclude that* $(x, y) \in i_A$.

Assume to the contrary that $x \neq y$.

Since R is total, for all $x, y \in A$ s.t $x \neq y$ we get $xRy \vee yRx$.

According to De-Morgan $\neg\big(\neg(xRy) \wedge \neg(yRx)\big)$.

From the definition of S we get $(x, y) \notin S$ Contradiction.

Therefore $x = y$,meaning $(x, y) \in i_A$.

Bu definition of containment we get $S \subseteq i_A$.

ב.

ג. *B. Disproof:*

Choose $A = \{1,2\}$ and define $R = \{(1,2), (2,2), (1,1)\}$.

We can easily see that R is reflexive, Anti-symmetric, and transitive, and in addition total on A.

By definition of S we get that $S = \emptyset$.

We get that S is not reflexive since $1 \in A$ *but (1,1)* $\notin S$

Question 12 (20 points)

Let A, B be sets. Prove or disprove:

1. $P(A) \cap P(B) = P(A \cap B)$
2. $P(A \cup B) = P(A) \cup P(B)$

1. *1. Proof (See lecture 4)*
2. *2. Disproof (See lecture 4)*

Discrete Mathematics - Formula Sheet for the Exam

Set theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$

$$A \times B = \{(a,b)|\ a \in A\ and\ b \in B\}$$

$$P(A) = \{S|S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$

$$|P(A)| = 2^{|A|}$$

De-Morgen laws:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad \mathrm{A}\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$

$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad \mathrm{A}\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive laws:

$$\mathrm{A} \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\mathrm{A} \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions:

$$i_{\mathrm{A}} = \{(a,a)|a \in A\}$$

Logics:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$

$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$

$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

$$\neg\forall \mathrm{x}(\alpha) \equiv \exists x(\neg\alpha)$$

$$\neg\exists \mathrm{x}(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k objects from n distinct objects:

	No repititions	With repititions
With order	$\frac{n!}{(n-k)!}$	n^k
No order	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$	$\binom{k+n-1}{n-1}$

Binomial formula:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \dots + i_k = n$$

Inclusion-Exclusion Principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$